Pitot-Static System (Mod 6/2033) – General Data

1. General

The pitot-static system provides the air pressure input for the Air Data, Attitude and Heading Reference Systems, ADAHRS 1 and ADAHRS 2, refer to 34-23-00, Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) – General Data and to the Electronic Standby Instrument System, ESIS, refer to 31-20-21, Electronic Standby Instrument System (ESIS) (Mod S.O.O. 6209, Mod 6/2049) – General data. Refer to Figure 1.

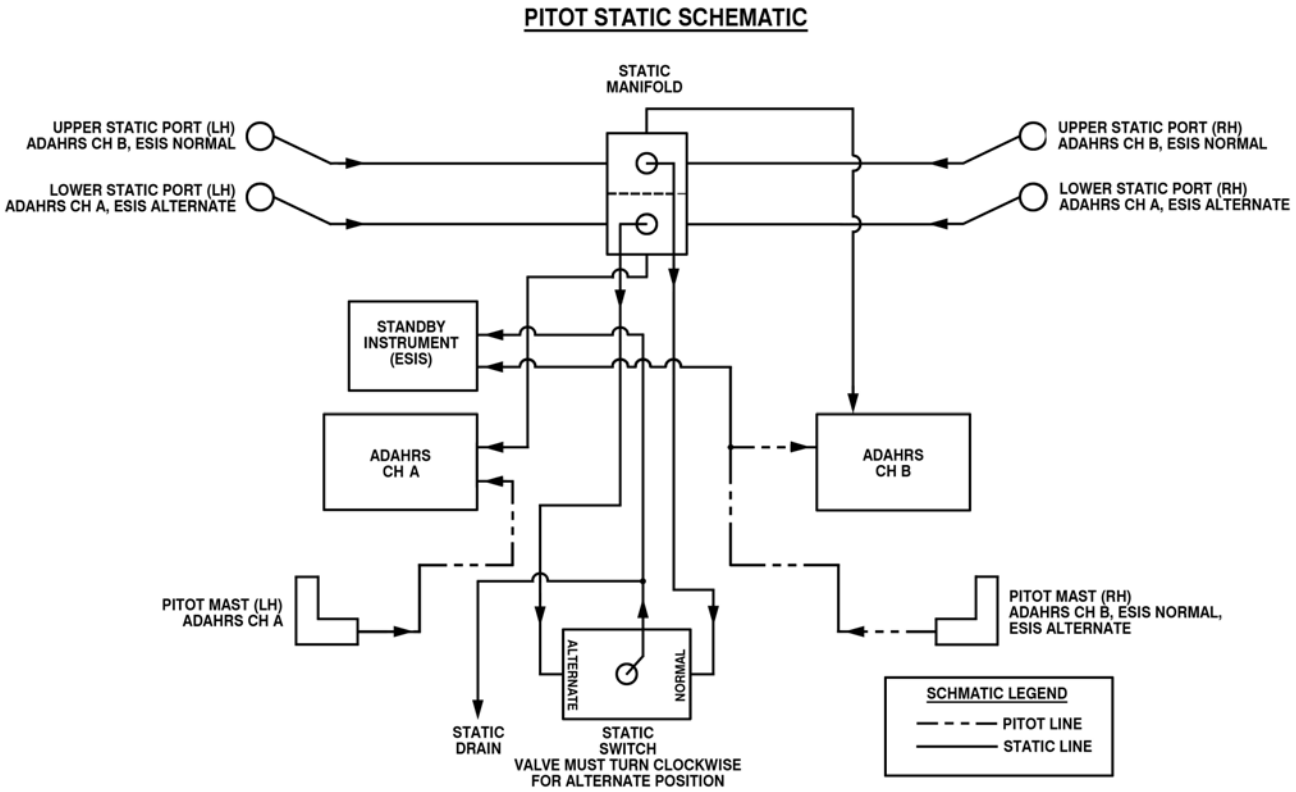
2. Description

A. Pitot Heads

There are two electrically heated pitot heads, each installed on a mast projecting from the left and right hand sides of the aircraft nose. The left hand pitot head supplies air pressure to the Air Data, Attitude and Heading Reference Systems, ADAHRS 1, the right pitot head supplies air pressure to for the Air Data, Attitude and Heading Reference Systems, ADAHRS 2 and to the Electronic Standby Instrument Systems, (ESIS).

B. (ESIS) Static Source Valve

The (ESIS) STATIC SOURCE valve is located on the dash panel, directly above the avionics circuit breakers. The valve can be selected to the NORMAL or ALTERNATE static source position. With the valve selected to the NORMAL position, the ESIS static lines will be connected to the upper static vents. With the valve selector in the ALTERNATE position the ESIS static lines will be connected to the lower static vents.



Pitot Static Schematic
Figure 1

3. System Service

A. Service Pitot Heads and Static Ports

1) Keep pitot heads and static ports clean and free from foreign matter. Do not use polishing compounds.

2) Ensure that drain holes in pitot heads are kept open.

3) Function test pitot heaters.

CAUTION **DO NOT LEAVE PITOT HEATERS SWITCHED ON FOR LONGER THAN ONE MINUTE. DO NOT BLOW INTO PITOT HEADS WHILE THEY ARE CONNECTED TO INSTRUMENTS.**

4. Adjustment/Test

A. Connecting Air Data Test Set

CAUTION **DO NOT REVERSE PITOT AND STATIC CONNECTIONS ON THE AIR DATA TEST SET, OR DAMAGE MAY OCCUR.**

Note Connection to the air data test set must be accomplished in accordance with the air data test set manufacturer's instructions, if available.

1) Ensure that the PITOT HEAT switch is off and the PITOT HEAT L (A6) and PITOT HEAT R (B6) circuit breakers, on the main circuit breaker panel, are pulled and fitted with safety clips.

2) Connect the PITOT port of the air data test set to the desired aircraft pitot head(s) (refer to Para 2. for pitot head definition).

3) Seal drain hole(s) on the back side of any connected pitot head(s).

4) Connect the STATIC port of the air data test set to the desired static port(s) on either side of aircraft (refer to Para 2. for static port definition).

5) Seal the appropriate static port(s) on the unused side of aircraft.

B. Disconnecting Air Data Test Set

1) Ensure the air data test set is vented to ambient conditions.

2) Disconnect the STATIC port of the air data test set from any connected static port(s).

3) Remove seals from any unused static port(s).

4) Disconnect the PITOT port of the air data test set from any connected pitot head(s).

5) Remove seals from any connected pitot head(s).

6) Remove safety clips and reset the PITOT HEAT L (A6) and PITOT HEAT R (B6) circuit breakers on the main circuit breaker panel.

C. Leak Tests

Note For the following tests, ensure only the listed pitot head or static port is connected to the air data test set, all others must remain vented to ambient conditions.

1) Pitot Leak Test – Channel A

a) Connect the air data test set to LH pitot head (refer to Para 4.A).

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait several minutes for the aircraft electrical and avionics systems to initialize.

d) Ensure the pilot's PFD is displaying ADAHRS Channel A information (default).

e) Adjust the air data test set to provide 150 knots on the pilot's Primary Flight Display (PFD) airspeed indication.

f) With the air data test set sealed, or in leak test mode, verify that the pilot's PFD airspeed indication does not drop more than 2 knots in one minute.

g) Adjust the air data test set to provide 35 knots on the pilot's PFD airspeed indication.

h) With the air data test set sealed, or in leak test mode, remove the seal from the LH pitot head drain, ensure airspeed on the pilot PFD blanks.

i) Vent the air data test set to ambient conditions.

j) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

k) Disconnect the air data test set from LH pitot head (refer to Para 4.B).

2) Pitot Leak Test – Channel B, ESIS

a) Connect the air data test set to RH pitot head (refer to Para 4.A).

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Set the ESIS POWER switch to ARM.

d) Wait several minutes for the aircraft electrical and avionics systems to initialize.

e) Ensure the co-pilot's PFD is displaying ADAHRS Channel B information (default).

f) Adjust the air data test set to provide 150 knots on the co-pilot's PFD airspeed indication.

g) Verify the airspeed is increasing on the ESIS and co-pilot's PFD.

h) With the air data test set sealed, or in leak test mode, verify that the co-pilot's PFD airspeed indication does not drop more than 2 knots in one minute.

i) Adjust the air data test set to provide 35 knots on the co-pilot's PFD airspeed indication.

j) With the air data test set sealed, or in leak test mode, remove the seal from the RH pitot head drain, ensure the airspeed on the co-pilot's PFD blanks.

k) Vent the air data test set to ambient conditions.

l) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

m) Set the ESIS POWER switch to OFF.

n) Disconnect the air data test set from RH pitot head (refer to Para 4.B).

3) Static Leak Test – Channel A, ESIS Alternate

a) Connect air data test set to lower static port (refer to Para 4.A).

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Set the ESIS POWER switch to ARM.

d) Wait several minutes for the aircraft electrical and avionics systems to initialize.

e) Ensure the pilot's PFD is displaying ADAHRS Channel A information (default).

f) Set the STANDBY INSTRUMENT STATIC SOURCE valve to ALTERNATE.

g) Press the BARO knob on the pilot's PFD controller and ESIS to set baro to 29.92 inHg (STD).

h) Adjust the air data test set to provide an altitude 1000 feet above the ambient elevation.

i) Verify altitude is increasing on the ESIS and pilot's PFD.

j) With the air data test set sealed, or in leak test mode, check that the pilot's PFD altimeter indication does not drop more than 50 feet in one minute.

k) Vent the air data test set to ambient conditions.

l) Return the STANDBY INSTRUMENT STATIC SOURCE valve to NORMAL.

m) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

n) Set the ESIS POWER switch to OFF.

o) Disconnect the air data test set from lower static port (refer to Para 4.B).

4) Static Leak Test – Channel B, ESIS Normal

a) Connect the air data test set to the upper static port (refer to Para 4.A).

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Set the ESIS POWER switch to ARM.

d) Wait several minutes for the aircraft electrical and avionics systems to initialize.

e) Ensure the co-pilot's PFD is displaying ADAHRS Channel B information (default).

f) Ensure the STANDBY INSTRUMENT STATIC SOURCE valve is set to NORMAL.

g) Press the BARO knob on the co-pilot's PFD controller and ESIS to set baro to 29.92 inHg (STD).

h) Adjust the air data test set to provide an altitude 1000 feet above the ambient elevation.

i) Verify altitude is increasing on the ESIS and co-pilot's PFD.

j) With the air data test set sealed, or in leak test mode, check that the co-pilot's PFD altimeter indication does not drop more than 50 feet in one minute.

k) Vent the air data test set to ambient conditions.

l) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

m) Set the ESIS POWER switch to OFF.

n) Disconnect air data test set from the upper static port (refer to Para 4.B).

D. Function Test Pitot Heater

WARNING **USE CAUTION WHEN TOUCHING THE PITOT HEADS WHILE THE SYSTEM IS OPERATIONAL AS THE PITOT HEADS QUICKLY BECOME HOT ENOUGH TO CAUSE SEVERE BURNS.**

(Refer to 30-30-00, Pitot Heating (Mod 6/2033) – General Data).

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Select the PITOT HEAT switch to ON.

CAUTION **DO NOT LEAVE THE PITOT HEATER SWITCHED ON FOR MORE THAN ONE MINUTE.**

3) Confirm that the pitot heads warm up.

4) Select PITOT HEAT switch to OFF.

5) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

5. Cleaning

A. Clean Pitot and Static Pressure Lines

1) Disconnect pitot and static lines from instruments and accessories. Cap instrument and accessory ports.

2) Remove, clean and dry all moisture traps.

3) Using pressure source of 25 psi nitrogen or clean dry air, blow through each pitot and static line (from inside of aircraft to outside) until clear of dust and moisture.

4) Reinstall moisture traps and reconnect lines to instruments.

5) Carry out pitot and static leak test. (Refer to Para 4.).

6. Removal/Installation

A. Removal and Installation of Pitot Static Components

1) Ensure the pitot static lines are correctly identified.

2) Cap lines and component ports.

3) Carry out leak test on the reconnection of the lines to components as per Para 4.

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